

Federated Multi-Corpora Master README (README.md)

Project: __NovÆxorpus(NÆX) &
__Lex-Novī-Æxentis-Copiæ

Motto: *Xçineribus, in-variis-nunquam-varius,
Novi-Æxentis-Copiæ, Vincent*

System Ground Truth: Æsc & Æyre • Living Truth &

Universal Memory: #d.u.m.b.a.s.s.

1. Executive Summary & Architecture Overview

This repository orchestrates a federated, multi-corpora agentic system operating across a tri-node asymmetric mesh: Node Alpha (Moto RAZR Ultra 2025 / Snapdragon 8 Elite), Node Beta (NVIDIA Jetson Orin Nano Super), and Node Gamma (Rubik Pi 3 Dragonwing). The architecture strictly enforces sensory immutability, zero-trust daemon sandboxing, decoupled background execution via loopback ADB, and sub-5ms local memory retrieval.

2. Federated Multi-Corpora Directory Layout

```
novae-xorpus/ (FEDERATED MASTER CORPUS ROOT & LIVING TRUTH)
├── 📄 MAP.md # Top-level human
navigation ontology & concept graph
├── 📊 master_manifest.jsonl # Federated SHA256 hash
table linking all child repos
├── 📖 README.md # Master System
Documentation & Beginner Runbook
|
├── 📁 aesop-xi/ # Core Ethical
Orchestration Protocol
|   ├── manifest.jsonl # Local tier catalog
|   ├── arbitration/ # Resource scheduling &
agent priority arbiters
|   ├── policies/ # Hard boundary rules &
security clearance tables
|   └── .incognito_red_sandbox/ # Hidden shadow
gatekeeper & isolated eval sandbox
|
├── 📁 novus-aexenti/ # Reasoning Engine &
Multi-Agent Intelligence
```

— manifest.jsonl	
— dual_agent_router/	# 0.8B Triage vs 4B/9B
Query routing contracts	
— mem0_episodic/	# Rolling in-session
habit keys & preferences	
— obl_static_protocols/	# Static ground-truth
queries over Postgres/MCP	
— vector_recall_engine.py	# Local semantic vector
index	
— reasoning_bank/	# Multi-model state
ledger & recovery flywheel	
— active_execution_paths.json	# Step-by-step
transaction ledger	
— failure_logs/	# Structured post-mortem
failure schemas	
— post_regression/	# Rollback snapshots
preventing capability drift	
— 📁 novaexopia/	# Native Daemons,
Runtimes & "The Claw"	
— manifest.jsonl	
— horizons-ui/	# APK 1: Chromium WebView
frontend & chat tiles	
— aesc/	# APK 2: Shell/Terminal
Daemon & ADB Loopback (5555)	
— aeyre/	# APK 3: Sensory Ingress,
Silero VAD, Moonshine, Kokoro	
— openwiki-tui-harness/	# Interactive TUI Help
Desk & AST Knowledge Graph	
— modular_harnesses/	# Execution harnesses
(Prime Agent RLM, ECC skills)	
— mcp_connectors/	# Node.js Filesystem,
SQLite, and Postgres MCP servers	
— 📁 vendor-corpora/	# Isolated Immutable
External Libraries	
— qualcomm-qairt-sdk/	# Dedicated Qualcomm NPU
/ FastRPC Knowledge Base	
— manifest.jsonl	
— http-specs/	# Hexagon v79 HTP
micro-architecture	
— headers/	# libQnnHtp.so &
/dev/adsprpc-smd ioctl contracts	
— quantization-guides/	# INT4/INT8 GEMM rules &
tensor layout bounds	
— google-android-platform/	# Android OS Internal
Protocols	
— manifest.jsonl	

```

|       |— system-services/           # Foreground Service,
LMK, and AIDL architecture
|       |— media-gaming-sdk/         # Frame capture &
low-latency audio pipelines
|       |— developer-options/       # Wireless Debugging &
loopback pairing specs
|
|— 📁 skills-and-capabilities/       # Shared Plugin &
Analytical Library
|   |— manifest.jsonl
|   |— code-review-graph/           # PyGraphify / Graphify
AST dependency mappings
|   |— notebook-lmpy/               # Deep corpus analytical
query scripts
|   |— obsidian-skills/             # Vault hygiene and
graph-sync definitions
|   |— early-trend-scraper/         # Web research &
telemetry intake daemons
|
|— 📁 data_vault/                   # Living Semantic Memory
(02_wiki_md/)
|   |— manifest.jsonl
|   |— concepts/                   # Atomic linked notes
(Zettelkasten)
|   |— architectures/              # System blueprints
connecting child repositories
|       |— entities/               # Hardware profiles,
register maps & device contracts

```

3. Subsystem Breakdown: Roles & Responsibilities

Repository / Layer	Physical Target	Primary Runtimes	Operational Role
aesop-xi	Universal Mesh	Python / Bash / JSON	Enforces ethical boundaries, split execution rules, and houses the isolated .incognito_red_sandbox/ gatekeeper.
novus-aexenti	Node Alpha / Node Beta	Python / SQLite / MCP	Hosts the 0.8B/9B model router, mem0 episodic habits, OB1 static ground truth, and the Reasoning

Repository / Layer	Physical Target	Primary Runtimes	Operational Role
			Bank ledger.
novaexopia	Node Alpha (Moto RAZR)	Kotlin / Java / C++ FastRPC	Runs the decoupled 3-APK native framework (Horizons UI, Æsc terminal daemon, Æyre media daemon) and DroidDesk.
vendor-corpora	Node Alpha / Node Beta	Markdown / C++ Headers	Isolated, immutable reference repositories for Qualcomm QAIRT SDK and Google Android internals.
skills-and-capabilities	All Nodes	Python / Node.js / CLI	Houses extracted procedural tools (cli/, wrappers/), PyGraphify code graphs, and NotebookLM pipelines.
data_vault	Node Alpha / Mesh Sync	Obsidian Markdown / TUI	Semantic knowledge base managed via OpenWiki CLI with bi-directional wikilinks and atomic concept cards.

4. The Lean On-Device Runtime Architecture

To eliminate background terminations by the Android Low Memory Killer (LMK) and maximize battery efficiency, the mobile runtime operates strictly on bare-metal native daemons:

- **GenieX (Qualcomm Hexagon v79 NPU):** Executes GGUF Q4_0 quantizations of Qwen 3.5 via the GGML Hexagon backend with zero-copy ASharedMemory FastRPC buffers. LocalAI is completely purged.
- **OmniRoute (Port 20128):** Runs under Node.js LTS as an intelligent local gateway. Conserves tokens by routing prompts dynamically across local GenieX weights and remote APIs based on prompt cache hit rates and rate limits.
- **Prime Agent (Prime Intellect):** Persistent Python REPL driving the Recursive Language Model (RLM) loop and the Continual Harness online adaptation framework.
- **Decoupled Native 3-APK Topology:**

- Horizons UI: Presentation shell, WebSocket client, and model picker.
- Æsc: Registered Accessibility Service / Assistant daemon executing shell commands via ADB Loopback (127.0.0.1:5555) with UID 2000 shell privileges.
- Æyre: Low-latency sensory ingress daemon running Silero VAD, Moonshine ONNX STT, and Kokoro TTS.

5. Dual Operational Modes (ECC vs. Prime Agent)

Operational Vector	Mode A: Terminal Developer Session	Mode B: Sovereign Edge Assistant
Primary Driver	Claude Code CLI (Desktop / Termux Terminal)	Local On-Device Weights (Snapdragon Hexagon NPU)
Governing Harness	ECC (Everything Claude Code)	Prime Agent (Prime Intellect)
Active Tool Suite	honey-crush, nexus-mapper, ecc-planner, Pocock skills	Native Python REPL tools, SQLite KV lookups, FastRPC NPU calls
Harness Interaction	Autonomous developer loop inside the Claude CLI.	Prime Agent invokes Claude Code CLI strictly as a discrete external tool. No nested wrapping.

6. Beginner Runbook & Step-by-Step Initialization

- Clone & Verify Submodules:**

```
git clone --recurse-submodules <repo-url> ~/novae-xorpus && cd ~/novae-xorpus
```
- Establish Local ADB Loopback (One-Time Pairing on Node Alpha):**

```
# Enable Wireless Debugging in Developer Options, then pair:
adb pair 127.0.0.1:<PAIRING_PORT> <PAIRING_CODE>
adb connect 127.0.0.1:5555
```
- Launch the NPU Hot-Swap & Inference Daemon:**

```
python3
~/novae-xorpus/04_skills_runtime/extracted_tools/wrappers/npu_manager.py
```
- Start OmniRoute Local Gateway:**

```
omniroute --port 20128 --config
~/novae-xorpus/novaexopia/modular_harnesses/omniroute_config.json
```
- Run Zero-Trust System Housekeeper & Audit Sweep:**

```
bash
~/novae-xorpus/04_skills_runtime/extracted_tools/cli/system_housekeeper.sh
```
- Open the Æsc Interactive Control Panel:**

```
bash
~/novae-xorpus/04_skills_runtime/extracted_tools/cli/agent_panel.s
h
```

7. Associated Canonical Master Specifications

For exhaustive technical contracts, register layouts, and schema definitions, refer to the following canonical documents:

- 00_DEFINITIVE_MASTER_SPECIFICATION_V3_COMPLETE.md — Apex system architecture, directory schema, tier access rules, and bash scaffolding.
- 01_SOVEREIGN_NODE_AND_APK_TOPOLOGY.md — Decoupled 3-APK framework, ADB loopback, and tri-node hardware specifications.
- 02_DUMBASS_UNIVERSAL_MEMORY_SPEC.md — Ubiquitous #d.u.m.b.a.s.s. memory engine, Mem0, OB1, SQLite, and Continual Harness.
- 03_DUAL_OPERATIONAL_HARNESS_AND_MCP_SPEC.md — Mode A (ECC) vs Mode B (Prime Agent), MCP infrastructure, and PyGraphify code graphs.
- 04_ON_DEVICE_INGESTION_AND_W5H_FRAMEWORK.md — W5+H matrix, Markdown-first ingestion, and atomic JSONL marker generation.